

Varsha Danduri

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EDUCATION

University of Delaware, Honors College

Newark, DE | May 2028

B.S. + M.S. in Computer Science, 4+1 Accelerated Program | GPA 3.86

SKILLS & CERTIFICATIONS

Languages: Python, C, C++, Java, JavaScript / TypeScript, SQL

AI / ML: Model training and evaluation, NLP (tokenization, named-entity recognition), classifiers on medical and conversational data, LLM APIs (Anthropic, OpenAI), CrewAI multi-agent pipelines, GPU acceleration

Platforms & Tools: Flask, Node.js, PostgreSQL, Snowflake, GitHub Actions, VS Code, Conda, Android Studio, Xcode, Neovim

Certifications: PCEP Certified Python (2024), Udemy Python, Udemy ML / AI

RESEARCH EXPERIENCE

Undergraduate Researcher, Healthy LAIfe Research Lab, University of Delaware

June 2025 - Present

- Built one of the first multimodal data pipelines on the widely used MIMIC-IV health database, integrating structured records, clinical text, medical images, and physiological signals into a single system for medical research. Open-sourced on GitHub with 300+ stars.
- Trained models on the pipeline's output for medical research tasks and built a web interface that lets researchers define cohorts and pull data efficiently.
- Co-authored a research paper currently under review; presented a scientific poster at UD Summer Scholars and was invited to present at the Naval Academy Science and Engineering Conference (November 2025).

Chatbot Safety & Risk-Deferral Model

Research | Python, Machine Learning, App Deployment

- Built a model, deployed in an app, that reads user responses in chatbot conversations and flags signs of mental-health crisis in real time.
- Designed a deferral layer: when risk is flagged, the chatbot steps back and the conversation is routed to safety resources and escalation instead of a generated reply, directly targeting a known failure mode of conversational AI.

EXPERIENCE

Agentic Software Engineer, University of Delaware

December 2025 - May 2026

- Designed and deployed a multi-step agentic workflow (Windmill) that runs the department's TA / RA recruiting pipeline end-to-end, plus a full-stack dashboard (React, Node.js, PostgreSQL) that staff use to manage every student worker.

Software Engineer, University of Delaware IT

February 2026 - Present

- Engineered a CI/CD pipeline (GitHub Actions, Jest) validating a jQuery migration tool used across the university's enterprise web applications.

PROJECTS

NLP Python Package for Medical Notes

C++, Python, GPU Acceleration

- Developed a Python package with a C++ core for processing medical notes: tokenization for training transformers and classifiers, GPU-accelerated text segmentation, and automatic disease named-entity recognition mapped to ICD codes.

Kitin, Agentic Codebase Security Dashboard

Startup, funded by University of Delaware | AI Agents, Cloudflare, React

- Built an autonomous AI agent that does contextual code-logic analysis to catch vulnerabilities (e.g., SQL injection) that line-by-line tools miss, paired with a fine-tuned LoRA model and a git-automation engine with a full audit trail.

Skin Disease Classification App

Machine Learning, Node.js, Android (Edge)

- Trained an image classification model on the DermNet dataset and deployed it in a web and Android app for early detection and awareness of skin conditions.

AWARDS & RECOGNITION

- Best Modeling, UD DataQuest 2026: strongest predictive model in a university-wide data science competition.